

Seminar on Game Analytics

Presentation: Toxicity Detection in Gaming Communities

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Q&A Session Notes

Q1: What is the target of toxicity detection?

Answer: toxicity detection targets harmful behaviors in gaming communities such as hate speech, threats, and harassment, covering voice chat, text chat, and even visual signals like bullet-hole patterns used to form offensive symbols.

Q2: What is the difference between MTL and BERT?

Answer: The presenter explained that MTL (Multi-Task Learning) is used for voice toxicity. It works with audio signals using WavLM and trains on multiple tasks at once to reduce game noise. BERT is a text-based model used in ToxBuster that reads chat history and understands conversation context to detect harassment in written messages. So basically one handles audio, the other handles text.

Q3: What are the meaning of results or what does these numbers represent?

Answer: the numbers are performance metrics like recall and accuracy. For example, ToxBuster improved recall by 57%, meaning it caught 57% more toxic messages than older methods. The RL sampling approach improved detection recall by 35% without needing to monitor every lobby.

Q4: What is the result after detecting a toxic person?

Answer: The presenter explained that the research mainly focuses on the detection side. What happens after, like muting, banning, or flagging for review, is handled by the game platform's moderation system and is outside the scope of these papers. It is actually mentioned as a gap in the research.

Q5: How does the model differ if something is toxic or not?

Answer: this is the limitation of the study but it differs according to the country. Cultural differences mean that what one community considers normal banter could be flagged as toxic in another, which makes it hard to build a universal model.

Q6: Which of the five papers do you think is the most useful in real games?

Answer: The presenter explained that the BERT-based ToxBuster paper is the most useful, since most toxicity in games still happens in text chat. Voice and visual detection are nice ideas but harder to actually use.

Q7: What are the future research directions in this field?

Answer: there are three main directions: running models on consoles and phones, supporting more languages without retraining, and combining voice, chat, and player behavior together for better detection.

Q8: How big does the dataset need to be to train a good model?

Answer: the GameTox dataset used 53,000 examples, which is a good size. In general, more data is better because toxic messages are rare compared to normal ones.

Q9: Can these models work in real-time during a match?

Answer: The presenter explained that yes for text chat, but voice is harder because it needs more computing power. That is why some papers only check the lobbies most likely to have toxic players instead of all of them.